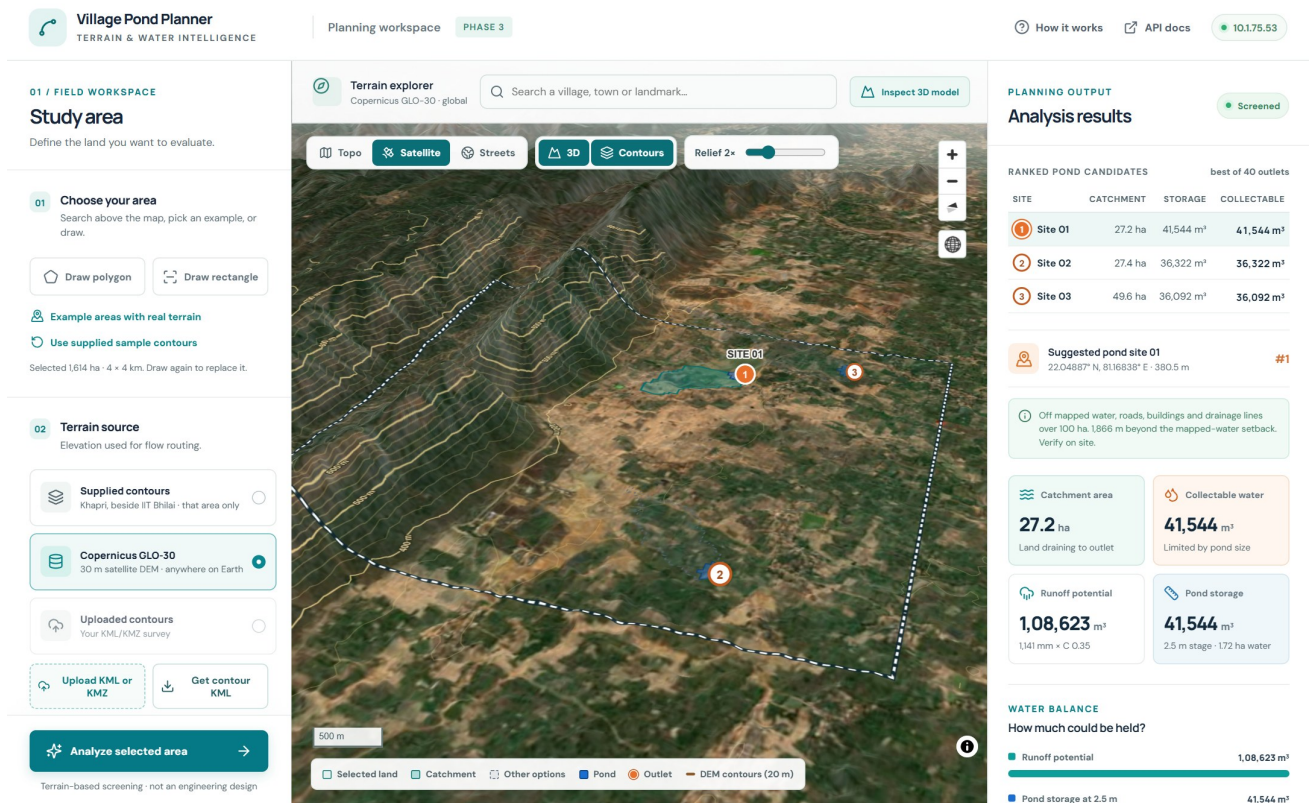




Live screenshot of the deployed planner: three ranked pond sites on the Maikal ridge front (Kabirdham), shown over Sentinel-2 imagery on 3D terrain.

AI-based Village Pond Planning System

An interactive web planner that finds pond sites, their catchments and the water they could hold, for any area on Earth

DEPLOYMENT

Live on the IIT network
verified 28 Sep 2026

VALIDATION

34 automated tests
pass locally and on the server

COVERAGE

11 verified areas
India and Ethiopia, 60°S–60°N

PLANNER

<http://10.1.75.53:3233/> (IIT Bhilai network or VPN)

API DOCS

<http://10.1.75.53:3233/docs>

PHASE 2 ROUTE

<http://10.1.75.53:3233/analyzeContour> (multipart field contour_map)

SOURCE CODE

<https://github.com/Galabavamsi/ai-village-pond-planning>

AI USE

Disclosed in Section 12

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1 Executive summary

Phase 2 delivered an API that turns an uploaded contour map into pond-site and catchment estimates. Phase 3 turns it into a complete planning tool that an administrator can use in a browser. They select land on a map, and the planner overlays three ranked pond sites. Each site comes with its upstream catchment, the water an embankment there could hold, and the runoff that a real rainfall record would deliver.

What Phase 3 delivers

- **Any area, real data.** The planner uses the supplied contour map, any uploaded KML/KMZ contour map, or the global Copernicus GLO-30 elevation model anywhere between 60°S and 60°N. Rainfall comes from CHIRPS v3 for a single month, the June–September monsoon or a full year. OpenStreetMap supplies water bodies, roads and buildings to avoid.
- **A physically consistent pond model.** Storage counts only land that drains to the pond outlet and lies below the embankment crest. Sites are ranked by the water they could actually collect. Each site reports a stage–storage curve, an indicative embankment length and warnings.
- **Earth view.** The map offers topographic, Sentinel-2 satellite and street basemaps, labelled contours, 3D terrain, a globe view, and a 3D model of the exact analysis grid with an optional satellite drape. Results export to Google Earth KML, GeoJSON and contour KML.
- **Deployed and verified.** The planner runs on the assigned IIT container. All 34 tests pass there. The browser test suites pass against the live address, from the campus network.

Before this final build, the Phase 3 prototype was reviewed line by line. The review found and fixed four defects that changed results:

- pond storage was inflated by 37–272 %;
- site ranking ignored storage;
- map screening failed for larger areas;
- the map lost the selected area when the terrain source changed.

A later check found and fixed a fifth: flat terraces created when coarse contours are interpolated. These defects are documented in Section 8.

Honest scope

All numbers are deterministic screening estimates computed from gridded elevation and historical rainfall. They are not an engineering design, a hydrological simulation or a land-eligibility decision. No language model computes any result. Seepage, evaporation, siltation, soil, spillway design and land ownership are not modelled, and the planner says so in every result.

2 Requirements and traceability

Phase 3 requirement	Implementation and evidence	Status
Select land on a map	Draw a polygon or rectangle (Terra Draw), pick one of 11 example areas, or search any place (Photon/OSM). The area and span are shown live and validated (0.25–10,000 ha, ≤ 20 km). Figs. 2, 3a.	Done
Analyze the land	POST /api/analyze-area with the supplied, uploaded or Copernicus elevation, CHIRPS or manual rainfall, and land-cover and pond settings (Section 5).	Done
Show pond location, catchment and water volume on the map	Numbered outlets, pond polygons and catchment polygons for all three sites (active one highlighted), plus runoff, storage and collectable volume, a stage-storage chart and benchmarks (Figs. 2, 6).	Done
Works beyond the sample	Copernicus GLO-30 anywhere at 60°S–60°N. Ten areas outside the sample were verified end to end (Section 7).	Done
No hard-coded results	All inputs come from the selected polygon and data source. The example list only moves the map. A test uploads and analyzes every shipped KML.	Done
Front end on the IIT system	http://10.1.75.53:3233/ on sys1 (SSH port 2233, Application 1 port), verified from the campus network (Section 10).	Done
Phase 2 API compatibility	POST /analyzeContour (field contour_map) and /findCatchment unchanged; Phase 2 tests pass.	Done
GitHub repository	https://github.com/Galabavamsi/ai-village-pond-planning , branch master.	Done
Report and demo video	This report; the video follows the walkthrough in Appendix C.	Report done

3 System architecture

The planner is one FastAPI application that serves both the API and the compiled React front end, so there is a single origin and a single process to run. Figure 1 shows the parts and the data they use.

Layer	Technology (version)
Front end	React 19, TypeScript 5.9, Vite 7, MapLibre GL JS 6.11 (globe, terrain, sky), Terra Draw 1.35, Three.js 0.186, lucide icons
API	FastAPI 0.141, Uvicorn, Pydantic request models, OpenAPI/Swagger at /docs
Geospatial	NumPy 2.5, SciPy 1.18 (interpolation, morphology), Shapely 2.1, rasterio 1.5 with GDAL 3.12, contourpy 1.4, pystac-client, planetary-computer
Testing	pytest (34 tests), Playwright with installed Chrome (desktop and mobile browser suites)
Hosting	Ubuntu 24.04 container on the IIT system, Python 3.12 virtual environment, tmux supervision

4 Features and user workflow

4.1 Choosing land anywhere

The workspace has three columns: the study-area controls, the map, and the results (Figure 2). A planner can:

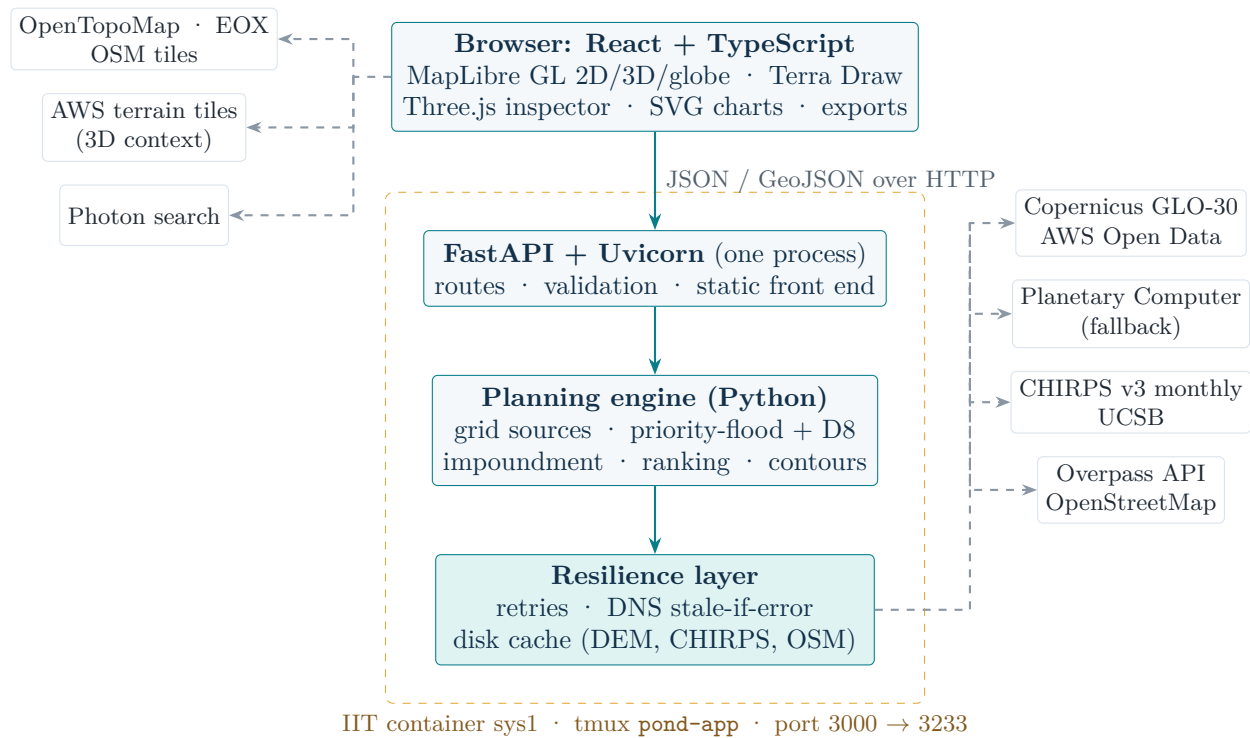


Figure 1: Architecture. Solid arrows are the planner’s own calls. Dashed arrows are remote data sources: the browser loads map tiles, terrain tiles and search itself, and the server reads elevation, rainfall and OSM features. Each server-side source is retried and cached.

- **draw** a polygon or rectangle on the map;
- **search** a village, town or landmark, which selects a 2 km square around it (Figure 3b);
- **pick an example area** with real terrain (Figure 3a).

The panel shows the selected area and its width and height in kilometres live, and rejects impossible sizes before any request is sent. Drawing outside the supplied or uploaded contours switches the terrain source to Copernicus GLO-30 automatically and explains why in a notice.

4.2 Terrain sources

- **Supplied contours:** the course KML (2,710 lines, 267–298 m). Section 6 explains its provenance.
- **Uploaded contours:** any KML or KMZ up to 20 MB. Each line’s elevation is read from its place-mark name, an `ExtendedData` field (`elevation`, `elev`, `z`, `height` or `level`), or a constant coordinate altitude. The uploaded linework is drawn on the map (Figure 4).
- **Copernicus GLO-30:** a 30 m global surface model, read for the selected area plus a buffer, so catchments can extend beyond the drawn land.
- **Get contour KML:** traces real contours for *any* selected area. The file opens in Google Earth and can be uploaded back as a contour survey.

4.3 Rainfall, land cover and pond settings

CHIRPS v3 provides a historical monthly rainfall total on a 0.05° grid. The planner can use one month, the June–September season (the Indian south-west monsoon, when most runoff occurs) or a calendar year. Only completed months are offered. Alternatively the user enters a rainfall depth. The runoff coefficient has land-cover presets within the ranges of the CGWB *Manual on Artificial Recharge* [8] (Table 1). The water stage (embankment height, 0.5–6 m) sets the pond size. An advanced setting caps the catchment treated as a “small pond” at 100 ha by default.

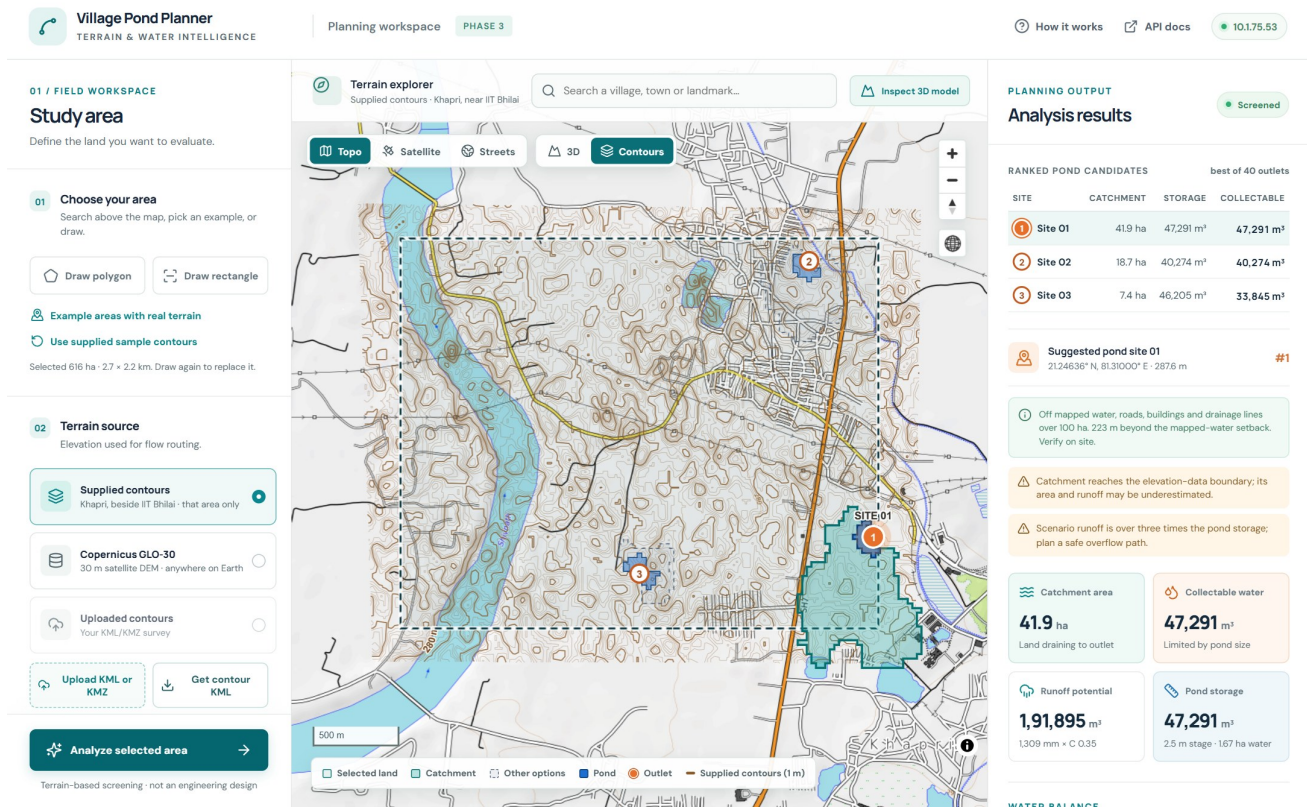


Figure 2: Supplied contour map (Khapri, beside IIT Bhilai) analyzed with the June–September 2025 CHIRPS rainfall. Three numbered outlets, their ponds (blue) and catchments (teal for the active site, dashed grey for the alternatives) sit over the labelled 1 m contours. The results column ranks the sites by collectable water and lists warnings. The header badge shows the live host 10.1.75.53.

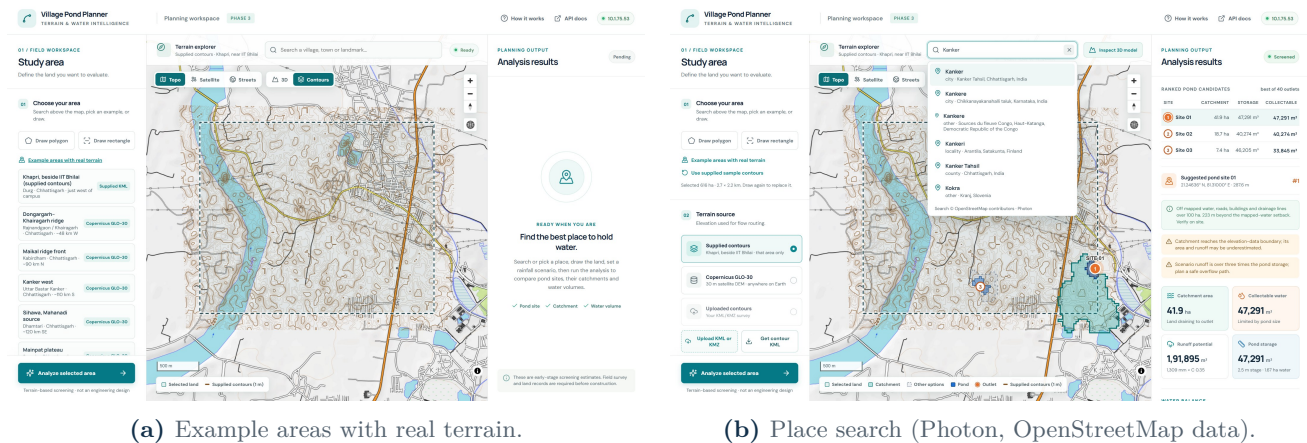


Figure 3: Two quick ways to reach an area other than the sample.

Preset	C	Basis (CGWB manual)
Forest	0.15	tree cover, volumetric range 0.05–0.2
Paddy	0.20	flat bunded cropland; Barlow 10–20 %
Scrub	0.25	grass, scrub and fallow land
Hilly	0.35	hills and plains with little cultivation; Barlow 35 %
Rocky	0.50	bare or rocky slopes, about 0.4–0.7
Built-up	0.75	roofs, roads, paved yards

Table 1: Runoff-coefficient presets. They are seasonal fractions for screening, not peak-flow design values.

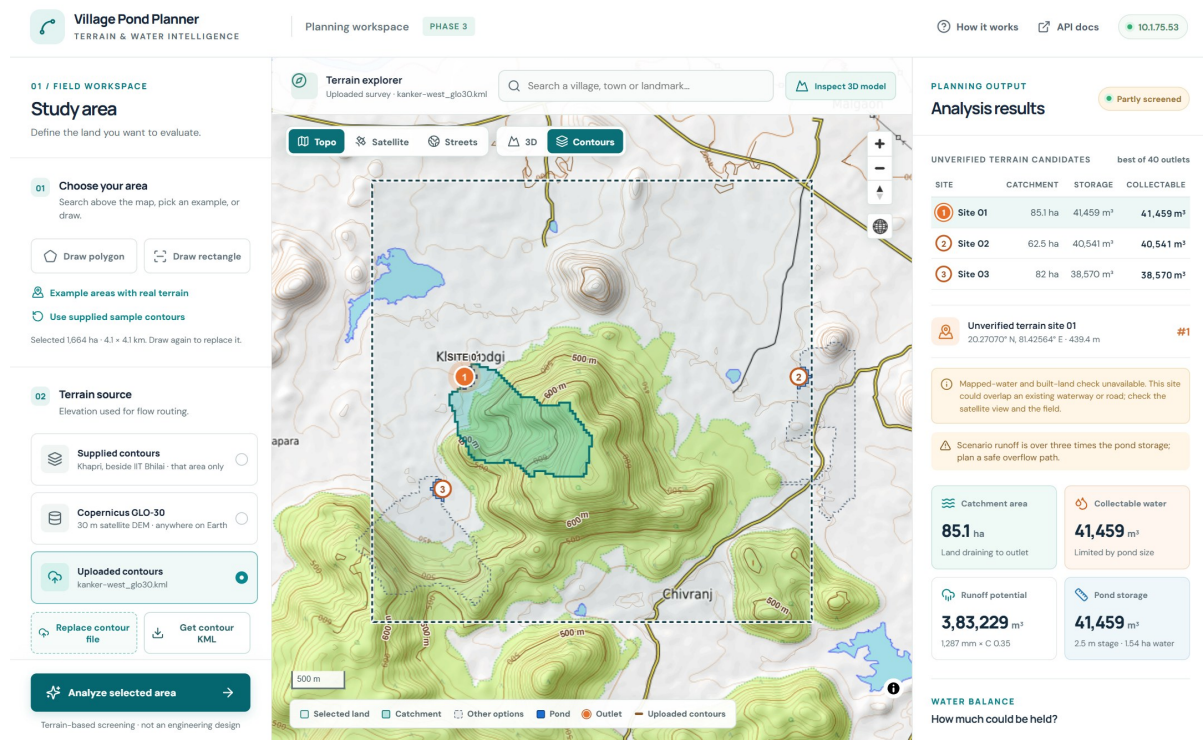


Figure 4: A real GLO-30-derived contour KML for Kanker west, uploaded and analyzed on the live server. The three outlets drain different valleys. During this run the server could not reach Overpass for this new area, so the result is honestly labelled “Partly screened” and “unverified terrain candidates”; it is not presented as cleared land.

4.4 Results, comparison and benchmarks

The results panel (Figure 6) supports the decision the administrator actually has to make:

- a **comparison table** of the three alternatives;
- **metrics** for the active site: catchment area, collectable water (and whether runoff or pond size limits it), runoff potential and pond storage;
- a **water balance** with bars on one shared scale;
- **pond facts**: crest level, mean depth, indicative bund length and storage per metre of bund;
- **benchmarks** against Mission Amrit Sarovar and MGNREGA farm-pond norms [6,7];
- a **stage-storage chart** with a crosshair tooltip and a table view;
- **monthly rainfall bars**;
- data **provenance**, and **exports**.

Each site also carries plain-language warnings: its catchment reaches the grid edge, it drains more than the small-pond cap, or runoff is more than three times, or less than half, its storage.

4.5 Maps, 3D terrain and globe view

The map toolbar switches between three basemaps: OpenTopoMap topography, EOX Sentinel-2 cloudless 2025 imagery and OpenStreetMap streets. It also toggles contours and **3D**, which tilts the map over global terrain tiles with adjustable relief. The globe button zooms out to the whole Earth with an atmosphere (Figure 7). Contours are labelled every fifth line. Before analysis they come from the survey; after a Copernicus analysis they are traced from the analysis grid. The map is created once and updated in place, so switching source or basemap never loses the selection.

01 / FIELD WORKSPACE

Study area

Define the land you want to evaluate.

01 Choose your area

Search above the map, pick an example, or draw.

Draw polygon

Draw rectangle

Example areas with real terrain

Use supplied sample contours

Selected 1,604 ha · 4 × 4 km. Draw again to replace it.

02 Terrain source

Elevation used for flow routing.

Supplied contours

Khapri, beside IIT Bhilai · that area only

Copernicus GLO-30

30 m satellite DEM · anywhere on Earth

Uploaded contours

Your KML/KMZ survey

Upload KML or KMZ

Get contour KML

Get contour KML

traces real contours for any selected area; the file opens in Google Earth and can be uploaded here.

03 Rainfall scenario

Water delivered to the pond.

CHIRPS v3 history

Enter rainfall

Jun-Sep

Full year

One month

Monsoon season of

2025

Sum of June–September CHIRPS totals; most Indian runoff falls then.

04 Land cover & pond

Runoff fraction and embankment height.

Forest 0.15

Paddy 0.2

Scrub 0.25

Hilly 0.35

Rocky 0.5

Built-up 0.75

Runoff coefficient

0.35

Fraction of rainfall that runs off the catchment.

Pond water stage

2.5 m

Embankment height above the outlet ground; sets pond size.

Advanced

Analyze selected area

Terrain-based screening · not an engineering design

Figure 5: The full study-area panel, split into two columns: 01 area selection with the live size check; 02 terrain source and the contour-KML export; 03 rainfall period and year; 04 land-cover presets, pond stage and the advanced catchment cap.

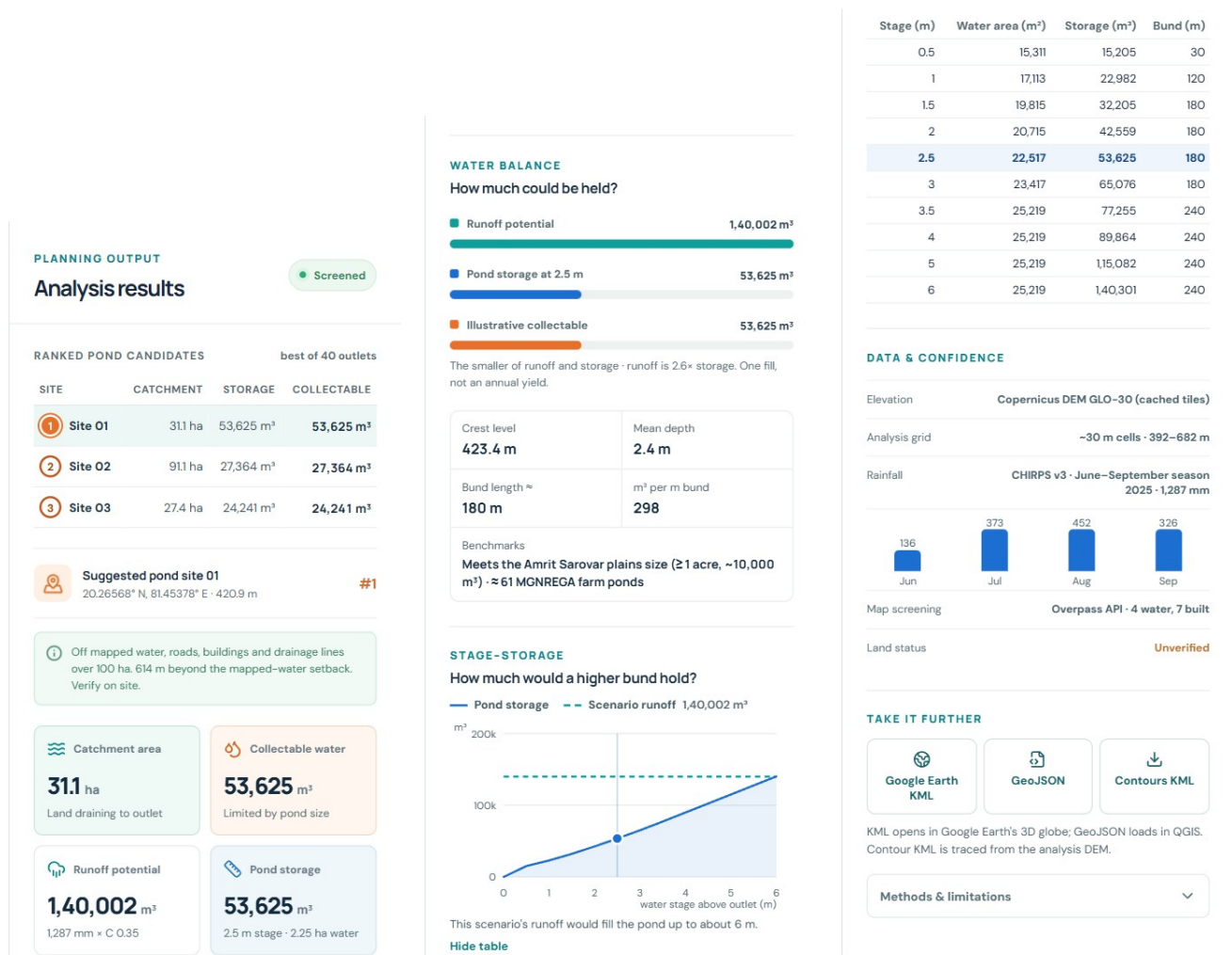


Figure 6: The complete results panel for Kanker west (live server), split into three columns. From left: ranked comparison and metrics; water balance, pond facts, benchmarks and the stage-storage chart; its table view, data provenance with the monthly CHIRPS bars, and exports.

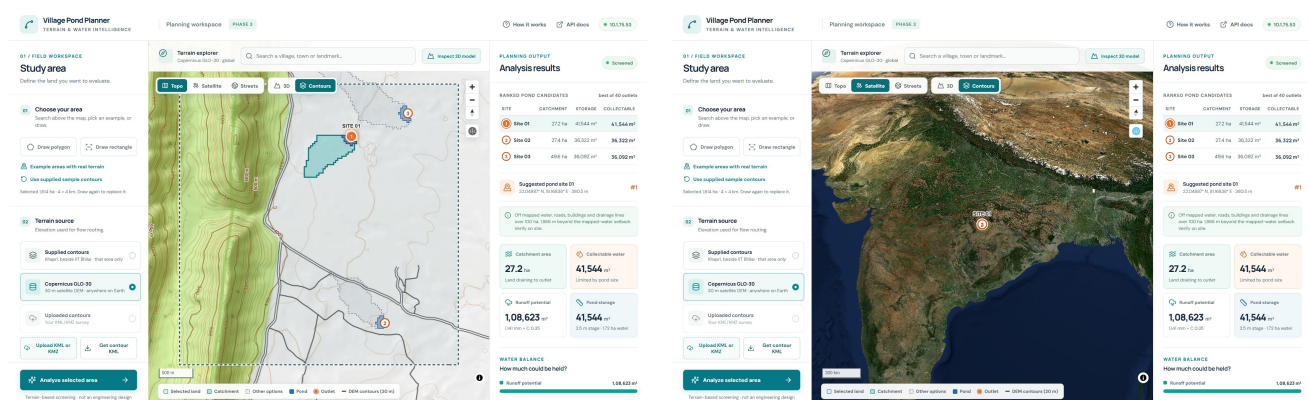


Figure 7: The same analysis on a 2D topographic map and on the globe. The cover shows it in tilted 3D over satellite imagery.

4.6 3D model of the analysis grid

Inspect 3D model opens a Three.js rendering of the *same* elevation grid used for routing (downsampled to at most 120×120), not a separate terrain service. It shows:

- a hypsometric colour ramp with contours draped on the surface;
- the study-area outline, the tinted catchment, the pond surface at crest level, and numbered outlets for all three sites.

The vertical exaggeration is set automatically so that relief spans about 12% of the model width (Equation 10); a slider adjusts it. **Satellite drape** textures the mesh with Sentinel-2 tiles (Figure 8). The exaggeration changes only the drawing, never the numbers.

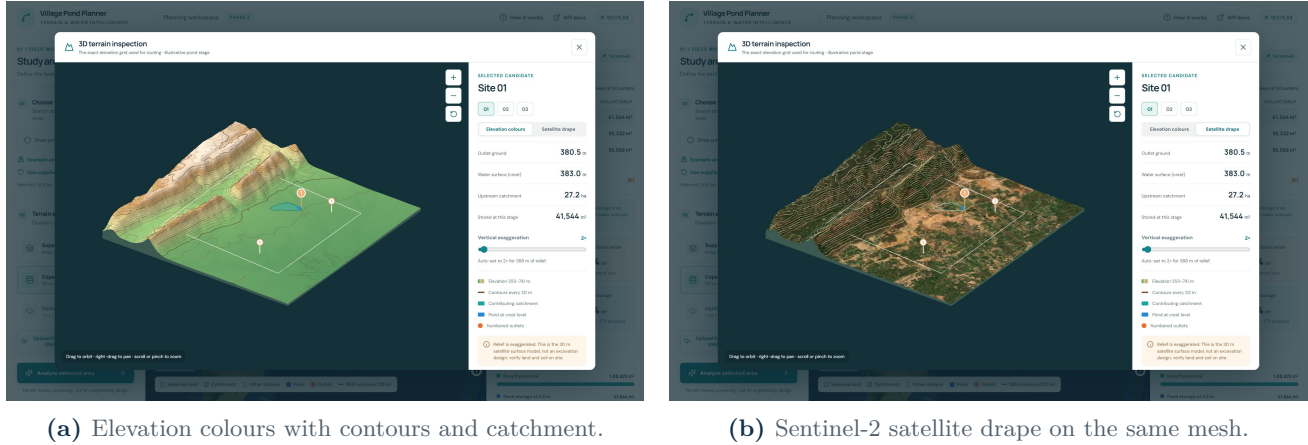


Figure 8: 3D inspector for the Maikal ridge front (live server).

4.7 Exports

- **Google Earth KML**: the study area, three catchments, ponds and outlets, styled and clamped to terrain, with every metric in each placemark’s description. It opens directly in Google Earth’s 3D globe.
- **GeoJSON**: the same features with numeric properties, for QGIS or other GIS tools.
- **Contour KML**: real contours for the selected area at a relief-based interval, with provenance and the Copernicus attribution in the file.

4.8 Mobile layout and accessibility

On phones the controls slide in from the side, and the map, results and a sticky **Analyze** button stack vertically (Figure 9). Every grey text colour was checked against WCAG AA. About 30 colours below a 4.5:1 contrast ratio were darkened, and the minimum text size was raised. The chart palette (teal runoff, blue storage, orange collectable) passed a colour-blindness separation check. Charts have legends and table views, so colour is never the only cue. The help dialog and 3D view close with Escape.

5 Algorithms

The analysis is deterministic. Given the same polygon, elevation source and settings it returns the same result, so results can be checked and reproduced. Algorithm version: `priority-flood-d8-impoundment-v4`.

5.1 Coordinates and grids

The polygon is validated: a simple polygon of at most 1,000 vertices, 0.25–10,000 ha, and no wider than 20 km. Longitude λ and latitude ϕ are projected to local metres around an origin (λ_0, ϕ_0) with $R = 6,371,000$ m:

$$x = R \cos \phi_0 (\lambda - \lambda_0), \quad y = R (\phi - \phi_0), \quad A_{\text{cell}} = |\Delta x \Delta y|. \quad (1)$$

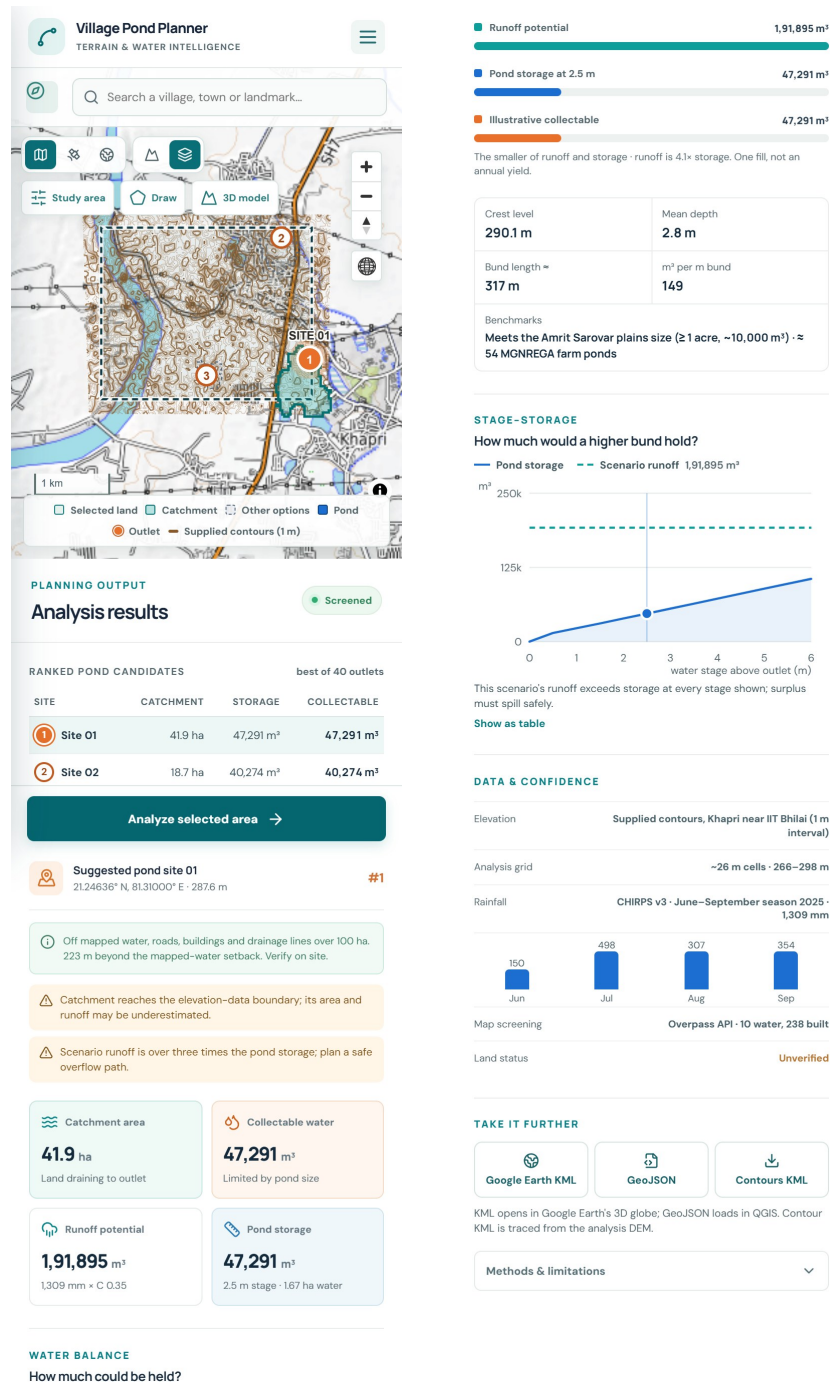


Figure 9: Mobile layout (390 px wide) on the live server, shown in two halves.

Contour sources. Contour vertices are interpolated to a regular grid of up to 220 cells along the longer side (the sample uses 125). A linear (triangulated) surface is flat wherever all three vertices of a triangle lie on one contour, which happens on hilltops and valley floors. With 20 m contours, 22.5 % of cells were flat, and a “pond” there became a flat patch whose storage was simply footprint $\times$ stage. Phase 3 therefore uses a C^1 Clough–Tocher surface z_{CT} , bounded by the contour interval Δ :

$$\tilde{z} = \text{clip}(z_{CT}, z_{\text{lin}} - \Delta, z_{\text{lin}} + \Delta). \quad (2)$$

Between two drawn contours the ground cannot cross another contour level, so this bound keeps the smooth surface consistent with the survey. It removes the overshoot that unbounded cubic interpolation produces (318 m on a survey whose top contour is 298 m). Outside the linework’s convex hull the nearest value is used. The Phase 2 route keeps linear interpolation, so its published results stay reproducible.

Copernicus source. The selected area’s bounding box is buffered by 30 % of its span, and by at least 0.009° (≈ 1 km). The 1° GLO-30 Cloud-Optimized GeoTIFFs that cover it are read from AWS Open Data and resampled bilinearly to at most 280×280 cells (≈ 30 m). Any gap is filled from Microsoft Planetary Computer’s copy. A box that crosses tile seams was tested and has no gaps.

5.2 Depression filling, D8 routing and accumulation

Depressions are filled with a priority flood from the grid edge. Cells are visited in order of spill elevation, and each unvisited neighbour n of the current cell c is raised to at least $z'_c + \varepsilon$ ($\varepsilon = 10^{-5}$ m). Flats therefore drain without cycles:

$$z'_n = \max(z_n, z'_c + \varepsilon). \quad (3)$$

Each cell i drains to the neighbour $n \in N_8(i)$ with the steepest downhill grade, where $d(i, n)$ is the orthogonal or diagonal distance:

$$f(i) = \arg \max_{n \in N_8(i), z'_n < z'_i} \frac{z'_i - z'_n}{d(i, n)}. \quad (4)$$

Contributing area accumulates along this graph in topological order:

$$A(i) = A_{\text{cell}} + \sum_{k: f(k)=i} A(k), \quad C(o) = \{i : \text{the flow path from } i \text{ reaches } o\}. \quad (5)$$

5.3 Screening

OpenStreetMap features for the area (plus a margin of about 200 m) are read from the Overpass API and turned into exclusion masks (Table 2). Each exclusion is inflated by half a cell diagonal, so that no retained cell footprint crosses a mapped boundary. A partial or timed-out Overpass answer is treated as a failure, because it could silently miss water; the result is then labelled *unverified*. Drainage lines with more than A_{max} of upstream area (default 100 ha), and their neighbours, are not used as small-pond outlets: a pond there would need an engineered spillway.

Mapped feature (OSM tags)	Exclusion buffer
Water bodies, wetlands, reservoirs, basins, water=*	40 m setback
Rivers and canals / streams, ditches and drains	45 m / 25 m
Buildings and built-up land use (residential, commercial, industrial, retail, military, cemetery)	12 m
Motorway, trunk / primary / secondary / tertiary	35 / 30 / 25 / 18 m
Unclassified, residential, road / service, living street	12 / 8 m
Tracks, paths, footways and other minor ways / railways	4 / 30 m

Table 2: Approximate setbacks. They are screening margins, not legal setbacks. The prototype used a flat 35 m for every road, including footpaths, which removed large amounts of village land.

5.4 Candidate outlets and the impoundment model

Candidate outlets are cells that:

- lie inside the selected land, off the exclusions and at least two cells from the grid edge;
- drain between $\max(1 \text{ ha}, 5A_{\text{cell}})$ and A_{max} .

Taken in order of contributing area, up to 40 are kept, each at least $\max(200 \text{ m}, 3 \text{ cells})$ apart.

For an outlet o at ground elevation z_o and water stage h , an embankment would hold water up to the crest $h_c = z_o + h$. The flooded set P contains every cell j that is connected to o by 4-neighbour steps within P and satisfies all of:

$$j \in C(o), \quad z_j \leq h_c, \quad j \text{ allowed}, \quad \|\mathbf{x}_j - \mathbf{x}_o\| \leq r, \quad (6)$$

where $r = \max(100 \text{ m}, 4 \text{ cells})$ keeps the pond at village scale. Storage, footprint and indicative embankment length are then

$$V_S = \sum_{j \in P} (h_c - z_j) A_{\text{cell}}, \quad A_P = |P| A_{\text{cell}}, \quad L_E = \sum_{\substack{(j,k): j \in P, k \notin C(o) \\ z_k \leq h_c}} \ell_{jk}, \quad (7)$$

where ℓ_{jk} is the length of the shared cell edge. Lower ground *outside* the catchment is where water would escape, so its edge length is the bund that has to be built. Repeating the calculation for $h = 0.5, 1, \dots, 6 \text{ m}$ gives the stage-storage table. The Phase 3 prototype flooded every connected low cell, including cells downstream of the outlet that a pond cannot hold. Section 8 shows the effect.

5.5 Rainfall, water volume and ranking

With rainfall depth P_{mm} (CHIRPS at the area centroid, or manual), catchment area A_C and runoff coefficient C :

$$V_R = \frac{P_{\text{mm}}}{1000} A_C C, \quad V_c = \min(V_R, V_S), \quad F = V_R/V_S. \quad (8)$$

V_c is the illustrative collectable volume for one fill, and F is the fill ratio. The 40 candidates are sorted by the key

$$(V_c \downarrow, V_S/L_E \downarrow, A_C \downarrow).$$

Three sites are then chosen so that their ponds never overlap and, where possible, no two share more than 60% of their catchments; this rule was added when two top sites turned out to be on the same drainage line. The three sites are alternatives, so their runoff is not added together.

5.6 Contours and 3D display

Contours are traced from the analysis grid by marching squares (contourpy) and simplified to a quarter of a cell. The interval is the smallest “nice” value at or above one twenty-fifth of the relief:

$$\Delta_c = \min\{v \in \{0.5, 1, 2, 2.5, 5, 10, 20, 25, 50, 100, 200\} : v \geq (z_{\text{max}} - z_{\text{min}})/25\}. \quad (9)$$

Every fifth line is marked as major. For a model of width S and relief Z , the 3D exaggeration is

$$e = \text{clamp}(\text{round}(0.12 S/Z), 1, 30). \quad (10)$$

This replaces the prototype’s fixed $15\times$, which made flat plains look like spikes.

5.7 Pseudocode

```
validate polygon (area, span, vertices)
grid <- supplied/uploaded contours (bounded smooth interpolation) or Copernicus GL0-30 (AWS, PC
  fallback, cached)
in parallel: OSM exclusions (Overpass, retried, cached); CHIRPS totals for the chosen months (retried,
  cached)
z' <- priority_flood(grid.z); f <- steepest_D8(z'); A <- accumulate(f)
allowed <- inside polygon AND NOT (mapped water OR built land)
```

```

outlets <- separated cells with 1 ha <= A <= A_max, allowed, not beside a major channel    (<= 40)
for each outlet o:
  C(o) <- upstream cells; P <- flood_fill(o, crest = z_o + h, within C(o), allowed, radius r)
  V_S, A_P, L_E, stage table; V_R = P_mm/1000 * A_C * C; V_c = min(V_R, V_S)
rank by (V_c, V_S/L_E, A_C); pick 3 with non-overlapping ponds and distinct catchments
return sites, GeoJSON geometries, contours, 3D preview, rainfall, screening status and limitations

```

5.8 Parameters

Parameter	Default	Where it is set
Water stage h	2.5 m (0.5–8)	slider, <code>stage_m</code>
Runoff coefficient C	0.35 (0–1]	presets and slider, <code>runoff_coefficient</code>
Small-pond catchment cap $A_{\max}$	100 ha (5–5,000)	advanced slider, <code>max_catchment_ha</code>
Rainfall	CHIRPS Jun–Sep 2025	<code>rainfall_period</code> , <code>rainfall_year</code>
Pond search radius r	max(100 m, 4 cells)	fixed
Minimum catchment	max(1 ha, 5 cells)	fixed
Outlet separation / candidates	max(200 m, 3 cells) / 40	fixed
Copernicus buffer / grid	30 % (≥ 1 km) / $\leq 280^2$ cells	fixed

6 Data sources, provenance and licences

Source	Used for	Licence and attribution
Copernicus DEM GLO-30 via AWS Open Data [2] (fallback: Planetary Computer)	Analysis elevation (surface model, EGM2008 heights), contour export	© DLR e.V. 2010–2014 and © Airbus Defence and Space GmbH 2014–2018, provided under COPERNICUS by the EU and ESA
CHIRPS v3 monthly [3]	Rainfall	CC BY 4.0; doi:10.15780/G2JQ0P
OpenStreetMap via Overpass [4]	Water and built-land screening	ODbL, © OpenStreetMap contributors
OpenTopoMap	Topographic basemap	CC-BY-SA
EOxCloudless 2025 [5]	Satellite basemap and 3D drape	CC BY-NC-SA 4.0: non-commercial academic use
OpenStreetMap tiles	Street basemap	OSMF tile usage policy
Terrain Tiles on AWS	MapLibre 3D terrain only	SRTM and GMTED2010 courtesy of USGS; ETOPO1 courtesy of NOAA
Photon (komoot)	Place search	OSM data; public server used within fair use

Two sources were deliberately rejected after checking their terms:

- **Esri World Imagery** requires an ArcGIS account for this kind of use.
- **The OpenStreetMap editing API**, which the prototype used, is reserved for editing, not read-only applications. It also refused a 14×12 km area.

Provenance of the supplied contour map

`contours_1m.kml` covers 81.281–81.313°E, 21.240–21.264°N. That is Khapri village farmland between the IIT Bhilai campus (centre about 81.318°E, 21.245°N) and the Shivnath river, and only a thin strip overlaps the campus. Its internal structure (a root `Folder` named `ContourMapGenerator` and lxml `py:pytype` attributes) matches exports from the free Contour Map Generator [9], which traces roughly 30 m satellite elevation. The 1 m contour interval is therefore finer than the real precision of the data. The planner labels it as supplied contours, not a field survey, and says so in its limitations.

Better sources for detailed design include:

- **Survey of India Open Series Maps:** free 1:50,000 GeoPDFs with 20 m contours, which need digitising [10];
- **ISRO Bhuvan CartoDEM v3:** 30 m, login required;
- **FABDEM:** bare-earth 30 m, non-commercial;
- **a village DGPS or drone survey** exported as contour KML.

Any DEM can be converted with `gdal_contour -f KML -i 5 -a elevation dem.tif out.kml`, and the planner reads the `elevation` attribute.

7 Verified example areas

Each area was run end to end with the final algorithm: GLO-30 elevation, the CHIRPS June–September 2025 total, Overpass screening, three ranked sites and contour export. Its contours are shipped in `contour-maps/real/`. The areas were chosen for real hill-to-plain relief with villages and seasonal drainage. The Ralegan Siddhi, Hiware Bazar, Alwar, Sukhomajri and Abreha we Atsbeha areas are known for community watershed work.

Area	Region	Relief (m)	Monsoon (mm)	Site 1 (ha)	Site 1 (m ³)
Khapri (supplied contours)	Durg, beside IIT Bhilai	266–298	1,309	41.9	47,291
Dongargarh–Khairagarh ridge	~48 km W	338–632	1,337	34.0	44,051
Maikal ridge front	Kabirdham, ~90 km N	353–741	1,141	27.2	41,544
Kanker west	~110 km S	392–682	1,287	31.1	53,625
Sihawa, Mahanadi source	Dhamtari, ~120 km SE	426–550	1,287	56.4	43,319
Mainpat plateau	Surguja, ~265 km NE	1,026–1,120	1,626	21.3	15,196
Ralegan Siddhi	Maharashtra	611–747	623	25.8	24,315
Hiware Bazar	Maharashtra	684–876	665	97.0	32,525
Bheekampura johads	Alwar, Rajasthan	393–634	1,130	17.2	39,658
Sukhomajri	Panchkula, Haryana	426–606	1,538	26.4	23,998
Abreha we Atsbeha	Tigray, Ethiopia	1,926–2,539	681	54.2	26,191

Table 3: Example areas (June–September 2025 CHIRPS, $C = 0.35$, 2.5 m stage). Site 1 is the best-ranked outlet; in every area it is limited by storage, not runoff. The Khapri row uses the supplied contours, so its relief is the grid’s, not GLO-30’s.

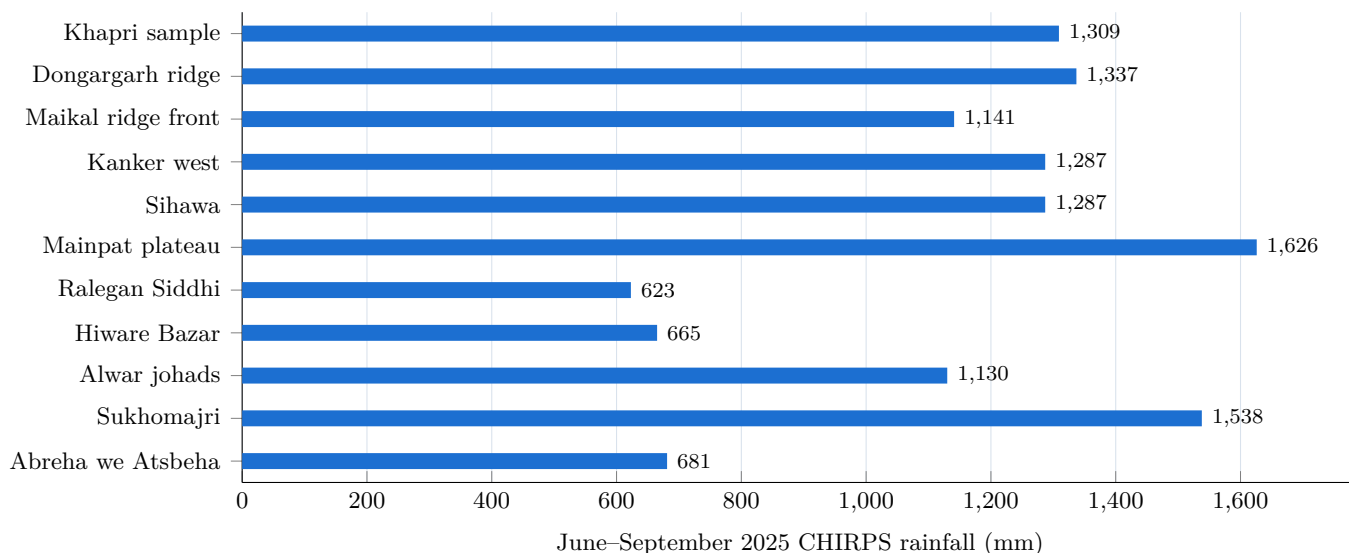


Figure 10: Monsoon rainfall read from CHIRPS for each example. The drought-prone Maharashtra villages receive half the rainfall of the Chhattisgarh sites. For comparison, the IMD 1991–2020 normal for Raipur is about 1,145 mm for June–September.

For Kanker west the stage–storage curve shows how a higher bund trades embankment length for water (Figure 11). At the default 2.5 m stage the pond holds 53,625 m³. The season’s runoff of about 140,000 m³ would fill it to about 6 m. That exceeds the Mission Amrit Sarovar plains benchmark (at least 1 acre and about 10,000 m³) and equals about 61 MGNREGA model farm ponds.

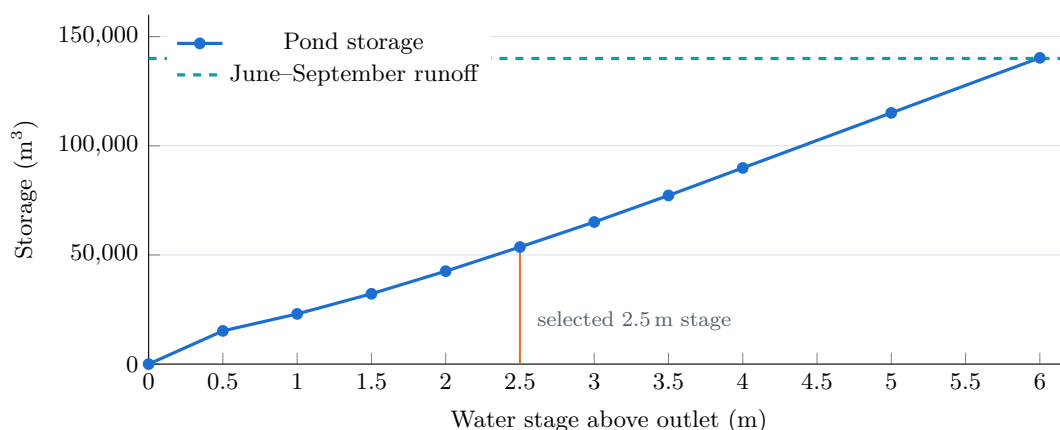


Figure 11: Stage–storage curve for the best Kanker west site (live result). Storage rises with stage while the embankment grows from 30 m to 240 m, so the curve helps choose a bund height.

8 Quality review: defects found and fixed

The Phase 3 prototype (26 September 2026) was reviewed with probe scripts, live network tests and browser screenshots. Every defect below was reproduced before it was fixed and is now covered by a test.

Area	Defect (measured)	Fix
Pond storage	Pond cells spread downstream and sideways outside the outlet's own catchment. That was 25–54 % of each footprint, and it inflated storage by 37–272 % (Figure 12a).	Impoundment model restricted to $C(o)$; the embankment length comes from lower ground outside it.
Ranking	Sites were the cells just below an arbitrary 98th-percentile “channel” cutoff. All three had 32–36 ha catchments on the sample (cutoff 36.3 ha), and storage never affected the rank.	40 separated candidates ranked by collectable volume; an explicit small-pond catchment cap.
Terraces	Linear interpolation of 20 m contours left 22.5 % of cells flat. Two sites shared one drainage line, and their storage was footprint $\times$ stage.	Bounded Clough–Tocher surface (0.6 % flat) and catchment-diversity rule.
OSM screening	The prototype used the editing API against its policy, and a 14 $\times$ 12 km area returned “unavailable”. It applied a 35 m buffer to every footpath.	Overpass with three servers, partial answers rejected, buffers by road class. The same area now screens 249 water and 15,361 built features.
Rainfall	A single month (August 2025) was treated as the planning rainfall.	Monsoon and annual CHIRPS totals fetched in parallel, with monthly bars.
Speed	DEM, OSM and CHIRPS were fetched one after another: 12.9 s for a Copernicus area.	Parallel fetches, a grid cache, AWS tiles: 6.3 s cold, 0.3 s repeat (Figure 12b).
Map	Switching the terrain source destroyed the map, so the drawn area vanished while the panel still said “Area selected”.	The map is created once and its layers are updated in place.
3D view	A fixed 15 $\times$ exaggeration made plains look like spikes, and the skirt dominated the view.	Automatic exaggeration, a thin skirt, draped contours, satellite drape.
Usability	The only way to reach another area was panning; drawing outside the survey just produced an error.	Search, examples, automatic source switching, a live area check.
Accessibility	About 30 text colours were below WCAG AA (e.g. #99a5a9 on white $\approx$ 2.6:1), and some text was 8–9 px.	Colours darkened to at least 4.5:1; minimum sizes raised.
Honesty	The sample was described as a campus survey, and the runoff presets had no source.	Provenance corrected; presets cite CGWB ranges.

9 Validation

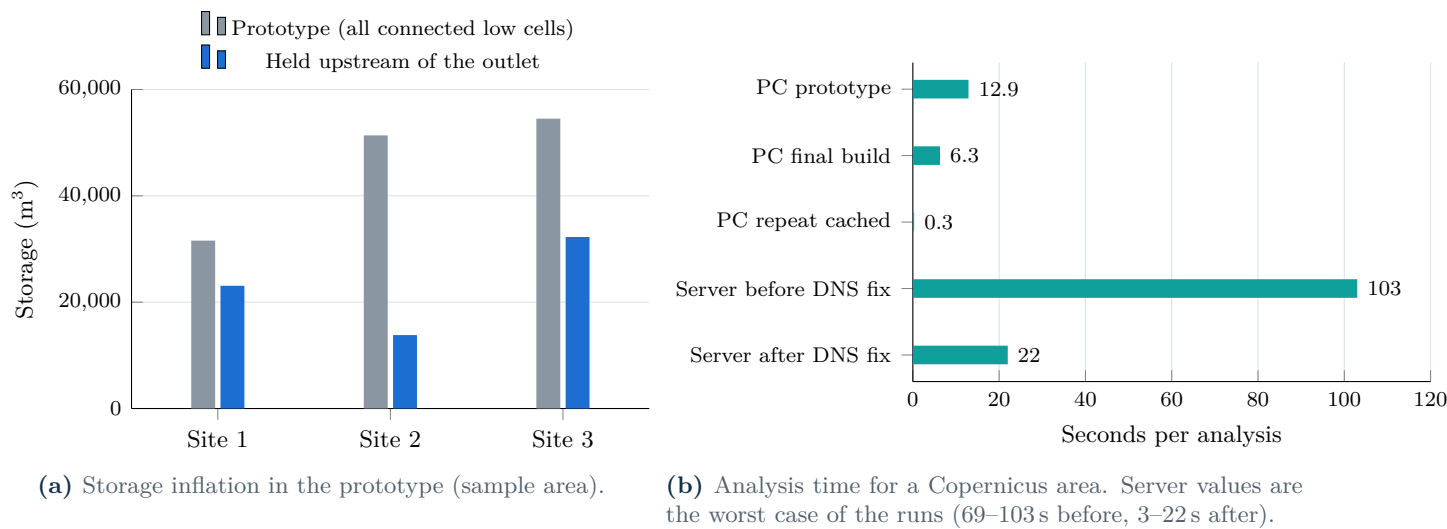


Figure 12: Two of the measured improvements.

9.1 Automated tests

Suite	Tests	What is checked
<code>test_api.py</code>	4	Phase 2 route: health, sample KML, KMZ, unsupported format
<code>test_planning.py</code>	6	Geometries and the runoff formula; different land gives different sites; outside-area rejection; contour overlay; sites and ponds avoid mapped water and roads; OSM XML parsing
<code>test_hydrology.py</code>	11	Pond inside its own catchment; ranking by collectable volume; consistent stage curves; non-overlapping ponds; higher stage holds more; contours in the response; a synthetic valley; rainfall periods; Overpass parser and road classes; contour export round trip
<code>test_upload.py</code>	8	Uploaded KML/KMZ drives the analysis; absolute altitudes and ExtendedData; every shipped real KML uploads and analyzes; bounded smooth interpolation; distinct drainage lines
<code>test_resilience.py</code>	5	AWS tile naming across hemispheres and seams; retries; disk cache; offline cached DEM; DNS stale-if-error

All 34 tests pass on the development PC and on the IIT server. Three Playwright browser suites drive a real Chrome, on desktop (1440×900) and mobile (390×844):

- `smoke.mjs`: analyze, draw a second area and confirm the site changes;
- `smoke-upload-3d.mjs`: upload two contour maps, open the 3D model, orbit it, and change site and exaggeration;
- `walkthrough.mjs`: satellite and 3D terrain, source switching, an example area, the inspector with satellite drape, the globe, and search.

Setting `POND_URL=http://10.1.75.53:3233/` runs them against the live deployment, where they pass with no page errors.

9.2 Live verification

Request on the deployed server (from the campus network)	HTTP	Time
GET /, /docs, /health, /api/config, front-end assets (JavaScript served as text/javascript)	200	<0.1 s
Sample analysis with the CHIRPS monsoon total	200	1.2 s
Cached example areas (Maikal ridge front, Abreha we Atsbeha)	200	0.7–1.1 s
KML upload (106 contour lines) / contour KML export	200	0.1 s
Phase 2 POST /analyzeContour with the supplied KML	200	2.9 s
New, uncached areas (AWS DEM, CHIRPS and Overpass fetched live)	200	3–22 s
Cached example with <i>all</i> network access blocked (proxy test)	200	0.4 s

10 Deployment on the IIT system

The application runs on the assigned container: SSH port 2233, so Application 1 is internal port 3000, published as external port 3233. It runs in tmux session `pond-app` under a loop script (`scripts/run_server.sh`) that restarts Uvicorn if it exits and logs to `logs/app.log`. Figure 13 shows the live API documentation.

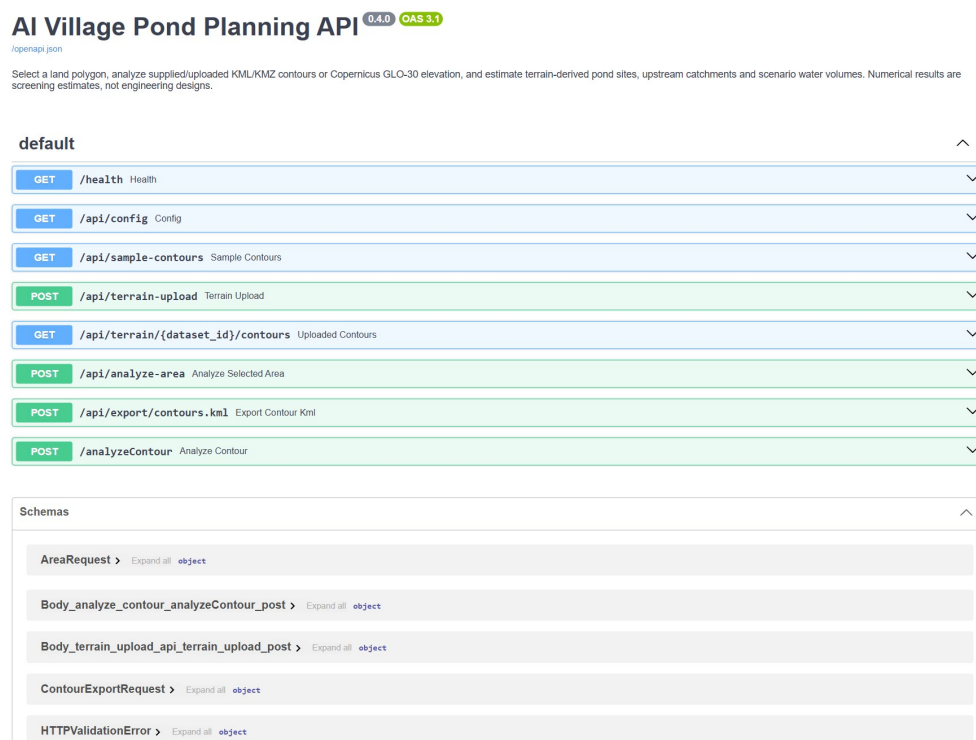


Figure 13: Swagger UI of the deployed API at <http://10.1.75.53:3233/docs>, listing the eight routes.

10.1 Constraints found and how they were handled

Constraint (measured)	Handling
No Node.js; PyPI downloads failed DNS or TLS checks	The front end is built on a workstation. Linux wheels (73 packages, 101 MB) are installed offline with <code>pip --no-index</code> . <code>uvloop</code> is added explicitly, because pip on Windows skips Linux-only dependencies.
SFTP restricted	Files are streamed through the SSH channel and verified by SHA-256.
Windows line endings broke the start script	<code>.gitattributes</code> keeps <code>*.sh</code> as LF.
Container DNS (1.1.1.1/8.8.8.8) drops lookups at random, about 20 s per failure	Retries on every remote read; an in-process DNS cache that reuses the last good address when a lookup fails; data hosts resolved at start-up; AWS as the primary DEM host. A new area went from 69–103 s with screening unavailable to 3–22 s fully screened.
Demo reliability	The 11 example areas are pre-cached (DEM, CHIRPS, OSM), so they need no outbound access.
From campus Wi-Fi (10.50.x), about 50 % of new TCP connections to 10.1.75.53 fail (10/20 on port 3233, 11/20 on port 2233), while 20/20 succeed from inside the server network	This loss is outside the application. The front end retries API calls after network-level failures, and the test scripts retry the first page load. Refreshing helps if the first load fails; wired LAN is recommended for the demonstration.

11 Limitations and responsible use

- **Elevation.** GLO-30 is a 30 m *surface* model, so trees and buildings raise it and it misses small channels and field bunds. Contour grids are interpolations. A catchment that reaches the grid edge is flagged as possibly underestimated.
- **Hydrology.** The runoff coefficient is a seasonal fraction, and CHIRPS pixels (≈ 5 km) are wider than a village. Collectable volume is one fill, not an annual yield. Seepage, evaporation (roughly 1.3–1.6 m a year near Raipur), siltation and spillway design are not modelled.
- **Screening.** OpenStreetMap is incomplete in rural Chhattisgarh. Forest land, sanctuaries and protected areas are not screened. Land ownership and legal eligibility are unverified; every site is labelled *unverified*.
- **Scale.** The pond radius cap keeps results at village-tank scale, so larger reservoirs are underestimated by design.
- **Imagery licence.** EOxCloudless imagery is licensed for non-commercial use only.

Every candidate needs a field visit, a topographic survey, soil tests and land records before any construction decision.

12 Disclosure of AI use

Generative AI was used extensively in the final Phase 3 build. This section states what it was used for, what it was not used for, and how its output was checked.

Tool	Claude Code (Anthropic), running the Claude Opus 5.5 model, used on 27–28 September 2026 on the development PC [12]. It ran two research sub-agents that used Firecrawl web search and scraping [13].
Code review and debugging	Reviewed the prototype for defects; wrote probe scripts that measured them (storage inflation, ranking clustering, failed OSM requests, the map rebuild, flat terraces); diagnosed deployment problems (restricted SFTP, CRLF endings, shell path conversion, missing Linux wheels, container DNS, campus packet loss).
Coding	Wrote most of the final Phase 3 code and tests: the impoundment and ranking model, bounded smooth interpolation, the Overpass screening, rainfall periods, the retry and cache layer, the DNS cache, contour tracing and export; and the front end’s search, examples, basemaps, 3D terrain and globe, results panel, charts, 3D inspector changes, exports and contrast fixes. Every commit from af86e61 to 79c2694 carries a Co-Authored-By: Claude trailer.
Research	Found and checked data sources, tile and API terms, the IIT Bhilai campus coordinates, the provenance of the supplied KML, example areas, and planning norms (Amrit Sarovar, MGNREGA, CGWB). Research claims were cross-checked by running the pipeline on each area and by reading the primary documents cited in the references.
Documentation	Drafted the README, HANDOVER.md, code comments and this report, including its figures, tables and equations.
Operations	Ran the tests, builds, browser suites, screenshot capture and deployment commands over SSH. It used credentials supplied by the team and asked for approval before destructive or public actions (cleaning the server, pushing to GitHub).

- **Not used for results.** The planner makes no calls to a language model. Every number in the application and in this report comes from deterministic code run on the cited data, or from measured test runs.
- **No generated images.** Every figure in this report is a screenshot of the running system or a chart or diagram drawn from measured values. The conceptual images in `output/imagegen/` come from earlier phases and are not used here as evidence.
- **Human responsibility.** The team set the goals and constraints, made the decisions (what to fix, what to clean up, when to deploy and publish), provided access, and reviewed the results. The team remains responsible for the submission.
- **AI errors that were caught.** Checking the AI’s output changed it in several places:
 - the sample was first described as the IIT Bhilai campus, and was corrected after the location was verified;
 - the first runoff presets had no source and were replaced by the CGWB ranges;
 - a first cubic interpolation overshot the survey by 20 m and was bounded;
 - a flat-terrace artefact was found by inspecting a live screenshot and was fixed before this report.

13 Future work

- A verified government-land parcel layer (e.g. Bhulekh or DILRMP data) as an eligibility mask, and forest and sanctuary boundaries.
- A bare-earth DEM (FABDEM or CartoDEM), and on-site drone or DGPS surveys, for design-grade storage.
- Monthly water-balance simulation with evaporation, seepage and demand, with a spillway check for large catchments.
- Land-cover-weighted runoff from ESA WorldCover instead of one coefficient.
- Durable, per-user storage of uploads (currently in memory for two hours) and a stable public mirror.

References and links

- [1] Project repository: <https://github.com/Galabavamsi/ai-village-pond-planning>
- [2] Copernicus DEM on AWS Open Data: <https://registry.opendata.aws/copernicus-dem/>; Planetary Computer: <https://planetarycomputer.microsoft.com/dataset/cop-dem-glo-30>
- [3] CHIRPS v3: <https://www.chc.ucsb.edu/data/chirps3>, doi:10.15780/G2JQ0P
- [4] Overpass API: https://wiki.openstreetmap.org/wiki/Overpass_API; OSM licence: <https://www.openstreetmap.org/copyright>
- [5] EOxCloudless: <https://cloudless.eox.at>; OpenTopoMap: <https://opentopomap.org/about>; AWS Terrain Tiles: <https://registry.opendata.aws/terrain-tiles/>; Photon: <https://photon.komoot.io>
- [6] Mission Amrit Sarovar, Phase-2 guidelines (MoRD, 2024): <https://amritsarovar.gov.in/AmrutSarovarDocuments/AmritSarovarGuidelinesPhase2.pdf>
- [7] MGNREGA suggestive estimates for farm ponds (MoRD circular, 2016): <https://nregaplus.nic.in/netnrega/writereaddata/Circulars/1569SuggestiveEstimatesFarmPondNADEPandVermiCompost.pdf>
- [8] CGWB, Manual on Artificial Recharge of Ground Water: <https://cgwb.gov.in/cgwbpm/public/uploads/documents/1679997242339591060file.pdf>
- [9] Contour Map Generator: <https://map.contourmapgenerator.com>
- [10] Survey of India Online Maps Portal: <https://onlinemaps.surveyofindia.gov.in>; Bhuvan: <https://bhuvan-app3.nrsc.gov.in/data/download/index.php>; FABDEM: <https://data.bris.ac.uk/data/dataset/s5hqmjcdj8yo2ibzi9b4ew3sn>; OpenTopography API: <https://portal.opentopography.org/apidocs/>
- [11] MapLibre GL JS: <https://maplibre.org>; Terra Draw: <https://github.com/JamesLMilner/terra-draw>; Three.js: <https://threejs.org>; FastAPI: <https://fastapi.tiangolo.com>; contourpy: <https://contourpy.readthedocs.io>; rasterio: <https://rasterio.readthedocs.io>
- [12] Claude Code: <https://claude.com/claude-code>
- [13] Firecrawl: <https://www.firecrawl.dev>

A API reference

Route	Purpose
POST /api/analyze-area	JSON: area (GeoJSON Polygon), source (sample upload copernicus), dataset_id, rainfall_source (chirps manual), rainfall_period (monsoon annual month), rainfall_year, rainfall_month, rainfall_mm, runoff_coefficient, stage_m, max_catchment_ha
POST /api/terrain-upload	Multipart contour_map (KML/KMZ, ≤20 MB) → temporary dataset_id, bounds, elevation range
GET /api/terrain/{id}/contours	Uploaded linework as GeoJSON
POST /api/export/contours.kml	Contour KML for area from the chosen source (optional interval_m)
GET /api/config, /api/sample-contours	Sample extent, defaults, example areas; supplied contours
POST /analyzeContour, /findCatchment	Phase 2 route (multipart contour_map)
GET /health, /docs	Status; Swagger UI

Errors use standard status codes: 400 for an empty upload, 413 for an upload that is too large, 415 for a wrong file type, and 422 for an invalid area, unusable contours or unavailable elevation. Each response recommendation contains:

- location, pond_region and catchment.geometry as GeoJSON;
- pond: storage, footprint, crest, mean and maximum depth, embankment length and stage_curve;
- water: runoff, collectable volume, fill ratio and limited_by;
- site_screening, notes, and land_status, which is always unverified.

The response also carries rainfall (with monthly_mm), water_screening, parameters, contours, terrain_preview and limitations.

```
curl -X POST http://10.1.75.53:3233/api/analyze-area -H "Content-Type: application/json" -d '{
  "area": {"type": "Polygon", "coordinates":
    [[[81.4178,20.2506],[81.4561,20.2506],[81.4561,20.2867],[81.4178,20.2867],[81.4178,20.2506]]]},
  "source": "copernicus", "rainfall_source": "chirps", "rainfall_period": "monsoon", "rainfall_year":
    2025,
  "runoff_coefficient": 0.35, "stage_m": 2.5 }'
```

B Reproduce and run

```
# Development (Windows PowerShell)
py -3 -m venv .venv; .\.venv\Scripts\python.exe -m pip install -r requirements.txt
cd frontend; npm ci; npm run build; cd ..
.\.venv\Scripts\python.exe -m pytest -q           # 34 tests
.\.venv\Scripts\python.exe run.py                 # http://127.0.0.1:8000/
cd frontend; node scripts\smoke.mjs; node scripts\smoke-upload-3d.mjs; node scripts\walkthrough.mjs
python scripts/prewarm_cache.py                   # fill cache/ for the example areas
python scripts/create_copernicus_contour_demo.py # regenerate contour-maps/real/*.kml

# IIT container (see README "Remote deployment with tmux")
python3 -m venv --without-pip .venv
.venv/bin/python ../wheelhouse/pip-*.whl/pip install --no-index --find-links ../wheelhouse pip
.venv/bin/pip install --no-index --find-links ../wheelhouse -r requirements.txt
tmux new-session -d -s pond-app "bash ~/ai-village-pond-planning/scripts/run_server.sh"
```

C Demonstration script

A walkthrough of under five minutes, in the order the browser test follows:

1. Open <http://10.1.75.53:3233/> and run the supplied contours; explain the three sites, catchments and warnings.
2. Open *Example areas* and choose *Maikal ridge front*; analyze.
3. Switch to Satellite and 3D, tilt the map, then zoom out on the globe.
4. Compare the sites in the table; read the stage-storage chart and the benchmarks.
5. Open *Inspect 3D model* and turn on the satellite drape.
6. Upload `contour-maps/real/kanker-west_glo30.kml` and analyze; export the Google Earth KML.
7. Close with the limitations: these are screening estimates that need a field survey.

D Repository map

Path	Contents
app/planning.py, hydrology.py	Area validation, grid sources, ranking; upstream index, impoundment, stage table
app/terrain.py, contours.py	KML/KMZ parsing and interpolation (Phase 2 algorithm); contour tracing and KML export
app/osm.py, rainfall.py, cache.py	Overpass screening; CHIRPS periods; retries, disk cache, DNS cache
app/main.py, examples.py, upload_store.py	Routes and models; curated examples; temporary uploads
frontend/src/	React app: App, MapCanvas, ResultsPanel, StageChart, PlaceSearch, TerrainInspector, exporters, api
contour-maps/, contour-maps/real/	Supplied KML, the original GLO-30 fixture, 10 example contour KMLs
tests/, frontend/scripts/	34 pytest tests; browser suites and the report screenshot script
scripts/	Contour generation, cache pre-warm, server start script
README.md, HANDOVER.md	Operating, deployment and handover documentation